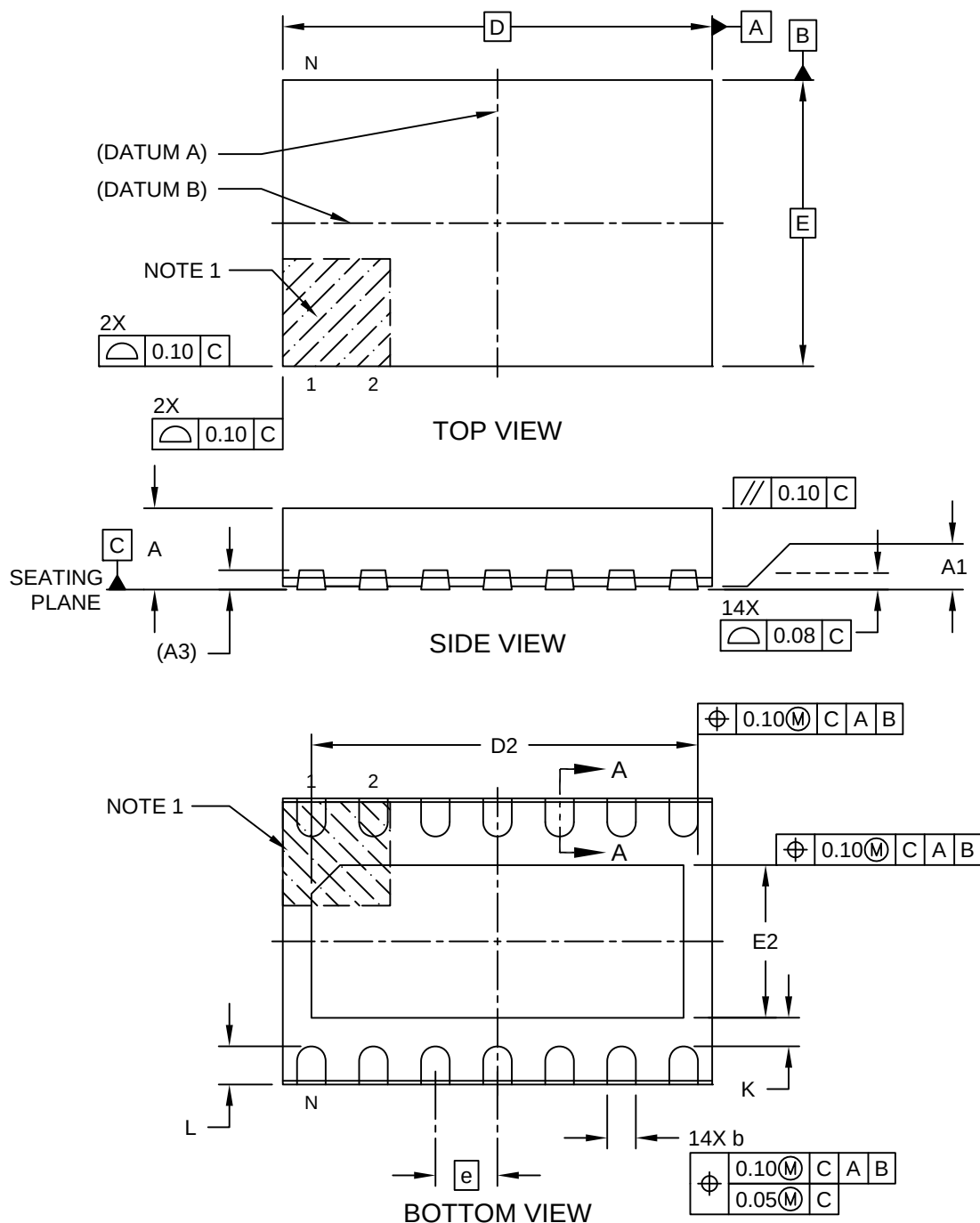


Packaging Diagrams and Parameters

14-Lead Very Thin Plastic Dual Flat, No Lead Package (QCB) - 4.5x3 mm Body [VDFN] With 1.6x3.9 mm Exposed Pad and Stepped Wettable Flanks

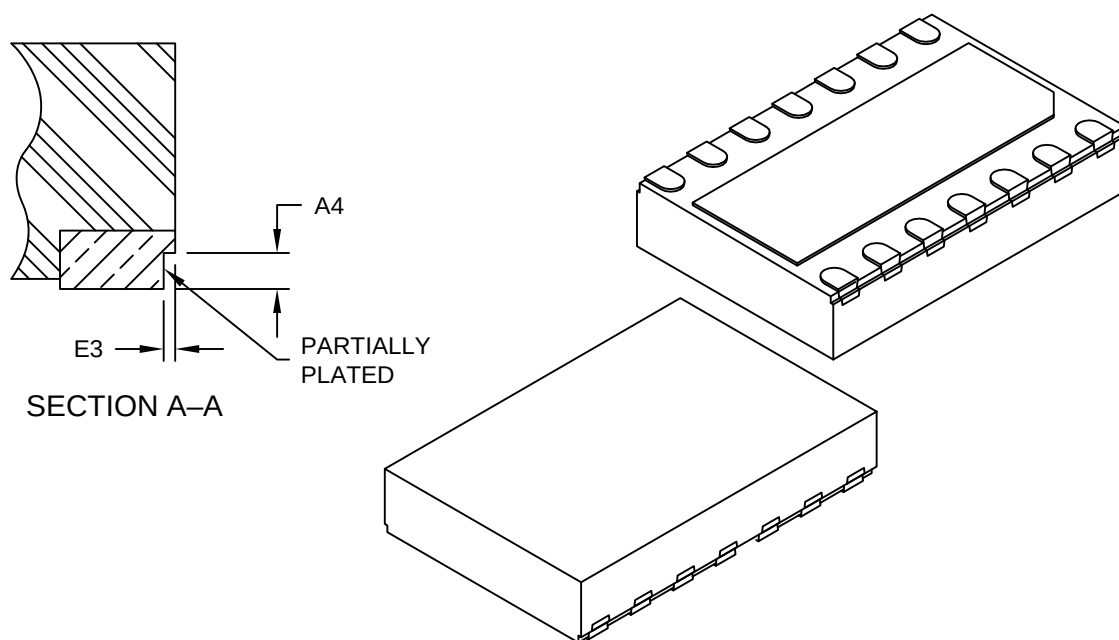
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		14		
Pitch	e		0.65 BSC		
Overall Height	A		0.80	0.85	0.90
Standoff	A1		0.00	0.03	0.05
Terminal Thickness	A3		0.203 REF		
Overall Length	D		4.50 BSC		
Exposed Pad Length	D2		3.80	3.90	4.00
Overall Width	E		3.00 BSC		
Exposed Pad Width	E2		1.50	1.60	1.70
Terminal Width	b		0.25	0.30	0.35
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.20	-	-
Wettable Flank Step Cut Depth	A4		0.10	0.13	0.15
Wettable Flank Step Cut Width	E3		-	-	0.04

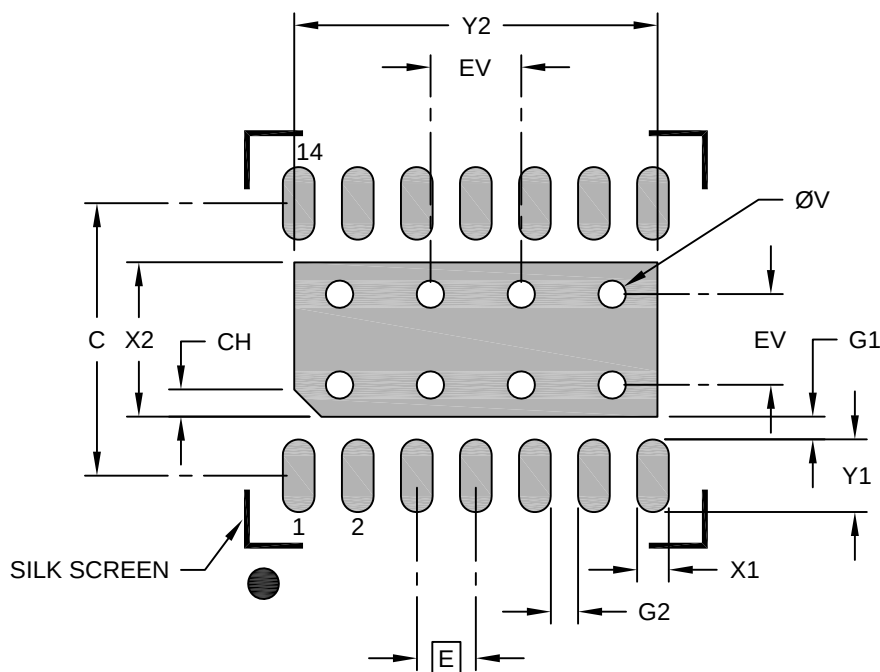
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

14-Lead Very Thin Plastic Dual Flat, No Lead Package (QCB) - 4.5x3 mm Body [VDFN] With 1.6x3.9 mm Exposed Pad and Stepped Wettable Flanks

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Optional Center Pad Width	X2			1.70
Optional Center Pad Length	Y2			4.00
Contact Pad Spacing	C		3.00	
Contact Pad Width (X14)	X1			0.35
Contact Pad Length (X14)	Y1			0.80
Pin 1 Index Chamfer	CH		0.30	
Contact Pad to Center Pad (X14)	G1	0.20		
Contact Pad to Center Pad (X12)	G2	0.20		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process